

MWANGAZA

Know which drought episodes may persist—before the next decision window closes.

Satellite-observed drought continuation
across IGAD

IGAD Hackathon 2026



A drought map tells us where stress is. Decisions need to know what may still be active next.

NOW

Risk level

Useful for situational awareness

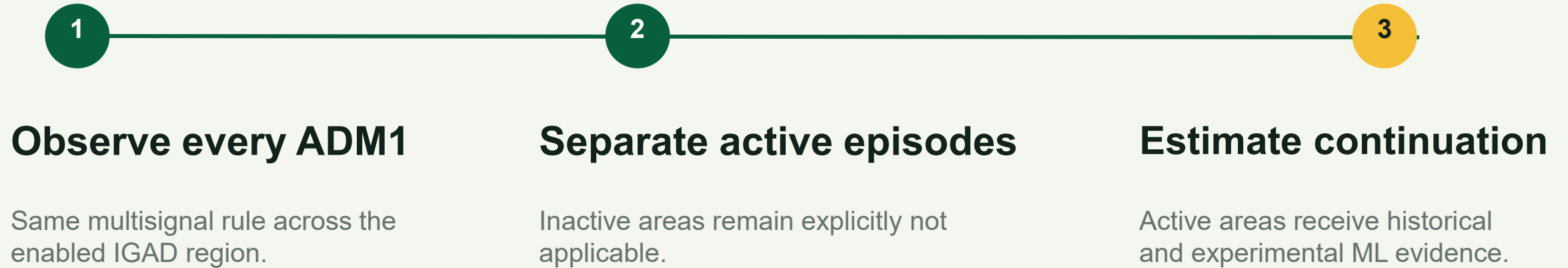
NEXT DECISION WINDOW

Will the same episode persist?

Useful for prepositioning, coordination and follow-up

Mwangaza makes that second question explicit at 30, 60, 90 and 180 days.

One operating flow, from regional scan to an actionable shortlist



The current materialization evaluates the full enabled IGAD footprint

8

countries

121

ADM1 evaluated

484

horizon results

11

active conditions

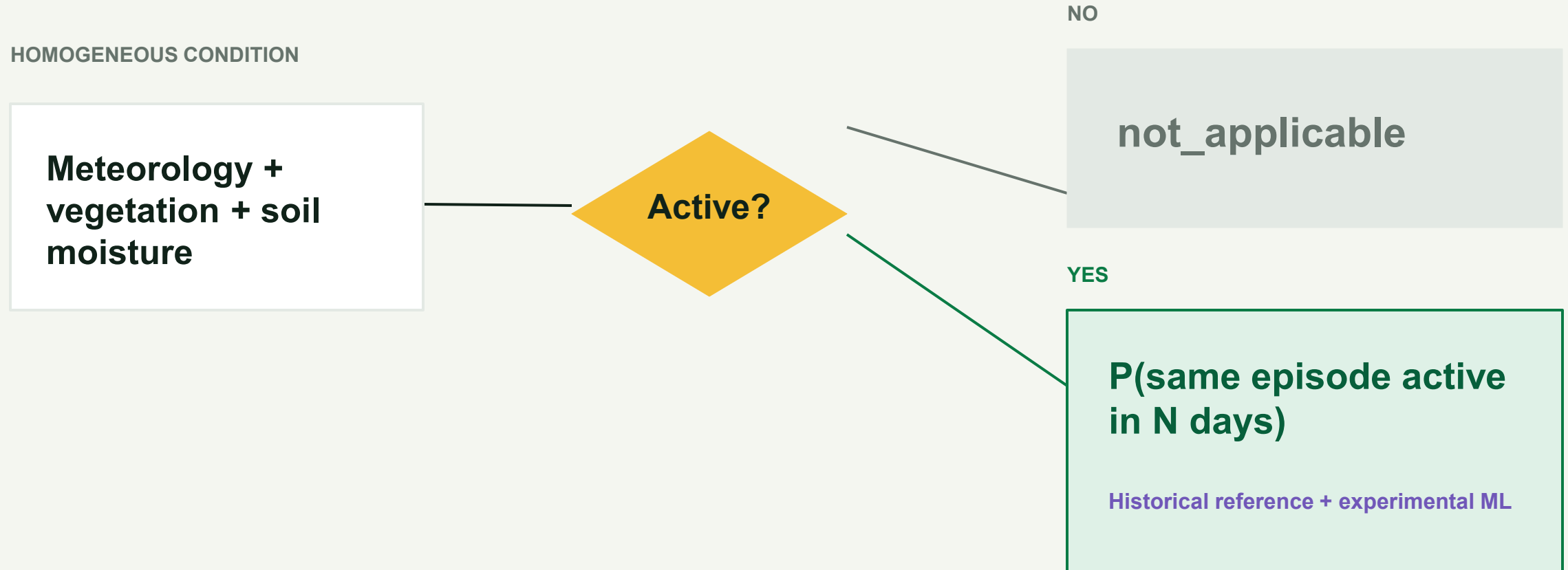
Analysis cut

20 Jul 2026

Each signal retains its own `observed_at`, `available_at`, `age_days` and `quality`.

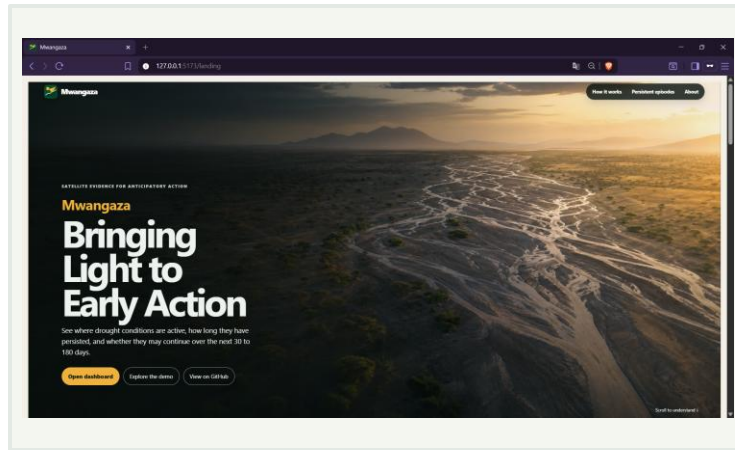
A probability appears only after the satellite evidence says an episode is active

HOMOGENEOUS CONDITION



The evaluator reaches the differentiator in three clicks

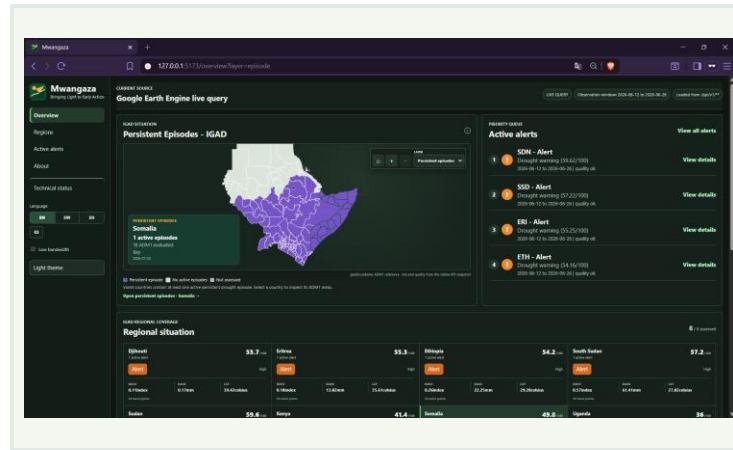
LIVE DEMO PATH



01 Landing

Promise and coverage

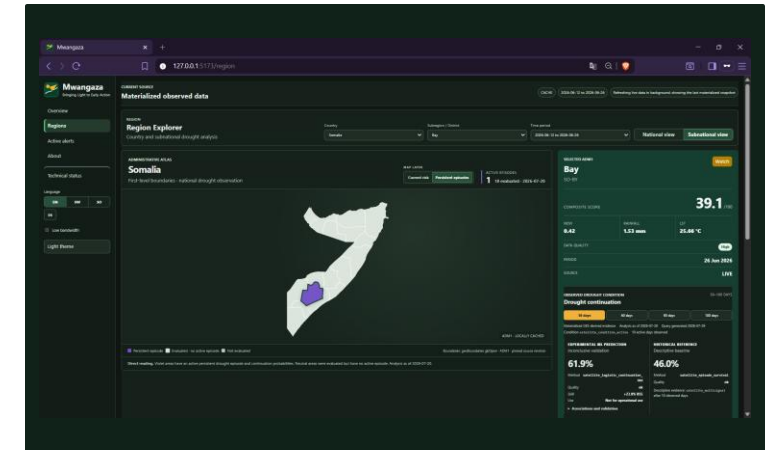
</landing>



02 Persistent episodes

IGAD active shortlist

</overview?layer=episodes>



03 ADM1 evidence

Horizons, probabilities and freshness

</region?layer=episodes>

The 30-day model adds signal in backtest—but the evidence is still too small for operational claims

+23.79%

Brier Skill Score vs historical
reference

0.0544

Expected Calibration Error

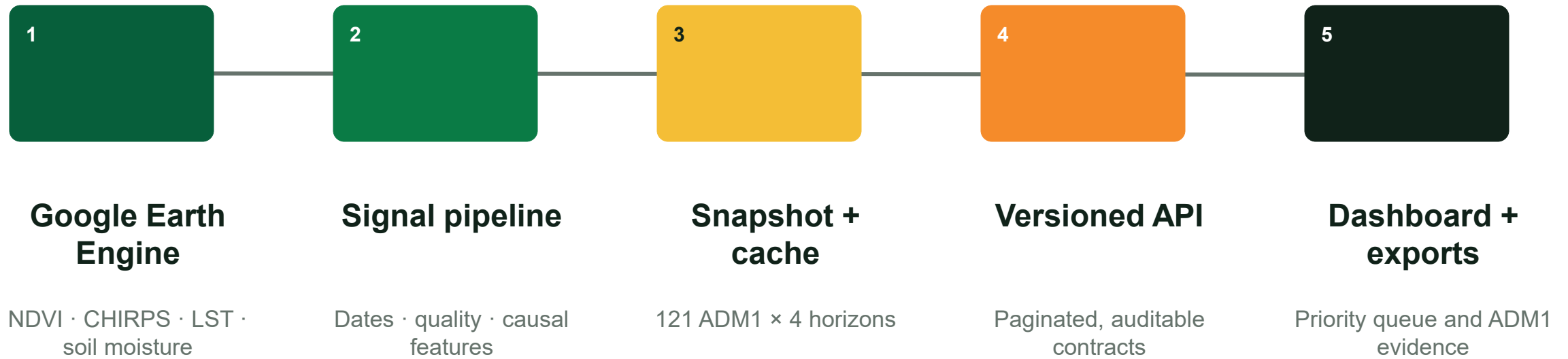
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walk-forward folds improved

EXPERIMENTAL · contract status: inconclusive · episode-weighted IC95 Δ Brier $[-0.0765, -0.0365]$

Backtest always hides future observations before estimating.

The live product serves materialized evidence; it does not run Earth Engine or training on request



Useful because the boundaries are explicit

MWANGAZA ESTIMATES

Whether the same satellite-observed drought condition remains active at a selected horizon.

IT DOES NOT ESTIMATE

Drought onset, exact duration, humanitarian impact or the action an operator must take.

NDMA

External official validation where available

FEWS NET

Impact evidence; IPC is not treated as a drought phase

NEXT STEP

Pilot the persistent-episode queue with regional drought analysts.

Validate priority decisions · collect new independent episodes · schedule reproducible refreshes

Mwangaza — Bringing Light to Early Action

